

create_pg_macro_conn_pattern

NAME

create_pg_macro_conn_pattern
Creates a macro power ground connection pattern. Use the set_pg_strategy command to instantiate the pattern and connect macros to the power ground network.

SYNTAX

status create_pg_macro_conn_pattern
pattern_name
-pin_conn_type long_pin | ring_pin | scattered_pin
[-nets net_list]
[-parameters var_list]
[-direction horizontal | vertical | @var]
[-width width | {hor_width ver_width} | @var]
[-layers layer_name | {hor_layer ver_layer} | @var]
[-spacing minimum | interleaving | spacing |
{hor_space ver_space} | @var]
[-pitch pitch | {hor_pitch ver_pitch} | @var]
[-number num | {hor_num ver_num} | @nvar]
[-excluded_pins pin_filter_spec]
[-via_rule via_rule_spec]
[-pin_layers layer_list | @var]

Data Types

pattern_name	string
net_list	list
var_list	list
var	string
width	float
hor_width	float
ver_width	float
layer_name	string
hor_layer	string
ver_layer	string
spacing	float
hor_space	float
ver_space	float
pitch	float
hor_pitch	float
ver_pitch	float
num	integer
hor_num	integer
ver_num	integer
nvar	string
pin_filter_spec	specification
via_rule_spec	specification
layer_list	list of string

ARGUMENTS

- pattern_name
Specifies the name of the macro pattern.
- pin_conn_type long_pin | ring_pin | scattered_pin
Specifies the macro pin and connection types. This is a required option and you must specify the long_pin, ring_pin or scattered_pin keyword. The long_pin keyword refers to long pins in the macro; orthogonal straps will be created for connection. The ring_pin keyword refers to internal ring pins inside the macro; finger connections will be created to connect the ring pin to a strap or external macro ring in the external power network. The scattered_pin keyword refers to the irregular pin shapes scattered inside the macro; extension straps will be created to connect scattered pins to the external power network.
- nets net_list
Specifies power and ground net names used in this macro connection pattern. The net names are symbolic names and will be mapped from net names specified by using real net names in the set_pg_strategy command.
- The net mapping is by position, that is, the i_th net name in set_pg_strategy command maps to the net name in the same position in this command. This option is not required and is needed when pin filtering based on net name is required. For example, use this option if certain nets can be skipped or pin connec-

tions in different nets have different via rules.

-parameters var_list

Specifies the list of parameters. Each parameter can be evaluated and used by other options in this command. This allows the pattern to be programmable and reused by passing different parameters in different situations. Each parameter var in the list var_list is a string and the at character (@) is used as a prefix to evaluate the parameter, such as @var. See the example section for more details.

-direction horizontal | vertical | @var

Specifies the direction of macro long pin connection straps. This option is required when you specify -pin_conn_type long_pin, otherwise, this option is ignored. This option specifies the direction for connection straps, not macro pins. The direction could be "horizontal", or "vertical", or specified by evaluating the variable var in the -parameters option. For parameters, the at character (@) is used as a prefix to evaluate the variable.

-width width | {hor_width ver_width} | @var

Specifies the width of the macro pin connection straps. If you specify -pin_conn_type long_pin, only one width value is required. Otherwise, the horizontal width and vertical width must be specified with hor_width and ver_width, respectively. The unit is defined in the technology file.

The width can also be specified by evaluating the variable var in -parameters option. The at character (@) is used as a prefix to evaluate the variable. By default, the minimum width defined in the technology file is used if this option is not specified.

-layers layer_name | {hor_layer ver_layer} | @var

Specifies the layer of the macro pin connection straps. This is a required option. If you specify -pin_conn_type long_pin, only one layer layer_name is required. Otherwise, the horizontal and vertical layers must be specified with hor_layer and ver_layer, respectively.

The layer names can be also specified by evaluating the variable var in -parameters option. The at character (@) is used as a prefix to evaluate the variable.

-spacing minimum | interleaving | value | {hor_space ver_space} | @var

Specifies the spacing between macro connection straps in the pattern. This option only applies if you specify -pin_conn_type long_pin or -pin_conn_type ring_pin. If you specify -pin_conn_type scattered_pin, this option is ignored.

minimum keyword refers to the minimum spacing between connection straps defined in the technology file, based on the wire width. interleaving keyword means that the distance between adjacent center lines of connection straps is equal.

Specific values can also be used for spacing. If you specify -pin_conn_type long_pin, only one value is required. If you specify -pin_conn_type ring_pin, the horizontal and vertical spacing values must be specified with hor_space and ver_space, respectively.

The spacing can also be specified by evaluating the variable var in -parameters option. The at character (@) is used as a prefix to evaluate the variable. By default, the minimum spacing is used if this option is not specified.

-pitch pitch | {hor_pitch ver_pitch} | @var

Specifies the pitch of macro connection straps in the pattern. The pitch only applies if you specify -pin_conn_type long_pin or -pin_conn_type ring_pin. If the connection type is scattered_pin, the pitch is ignored.

If the connection type is long_pin or ring_pin, at least one of the -pitch or -number options must be specified. If both options are specified, both options are honored. Otherwise, one value is derived from another by evenly distributing the connection straps in the macro.

If the pin connection type is long_pin, one value must be specified for the pitch value. If the type is ring_pin, horizontal and vertical pitch values must be specified by hor_pitch and ver_pitch, respectively. The unit is defined in the technology

file. The pitch values can also be specified by evaluating the variable `var` in the `-parameters` option. The `@` character is used as a prefix to evaluate the variable.

`-number num | {hor_num ver_num} | @var`

Specifies the repeating number of macro connection straps in the pattern. It only applies if you specified `-pin_conn_type long_pin` or `-pin_conn_type ring_pin`. If the connection type is `scattered_pin`, the number is ignored.

If the connection type is `long_pin` or `ring_pin`, at least one of `-pitch` or `-number` option must be specified. If both are specified, both are honored. Otherwise, one value is derived from another by evenly distributing the connection straps in the macro.

If the pin connection type is `long_pin`, one `num` must be specified for the repeating pattern. If the type is `ring_pin`, horizontal and vertical numbers must be specified by `hor_num` and `ver_num`, respectively. The repeating number can also be specified by evaluating the variable `var` in `-parameters` option. The `@` character is used as a prefix to evaluate the variable.

`-excluded_pins pin_filter_spec`

This option is not required. This option only applies when you specify `-pin_conn_type scattered_pin`.

For `scattered_pin` connection type, certain pin shapes that satisfy the `pin_filter_spec` specification can be excluded for connection. The `pin_filter_spec` specification uses the following format:

```
{layers : layer_list}
```

where `layers` is the keyword for pin layers. `layer_list` is a list of strings which contains a list of layer names. The pin shapes which satisfy the above criteria are excluded from connection. By default, all pin shapes are connected if this option is not specified.

`-via_rule via_rule_spec`

Specifies the via rule `via_rule_spec` between connection straps and macro pins in this macro connection pattern. The via rule defines the intersection locations and the via at the locations.

There are several ways to define the intersection locations in the specification `via_rule_spec`, where the syntax is defined as follows:

```
{intersection : all}      {via_def} |  
{intersection : adjacent} {via_def} |  
list_of_filter_rules |
```

The first method is `"{intersection : all}"`, where `intersection` is the keyword and `all` refers to all intersections between any two connection strap and macro pin across any layers.

The second method is `"{intersection : adjacent}"`, where `intersection` is the keyword and `adjacent` refers to all intersections between any two connection strap and macro pin only in adjacent layers.

With the intersection definition, the via specification `{via_def}` uses the following format:

```
{via_master : via_master_list}
```

where `via_master` is the keyword for via masters including via contact code defined in the technology file or user-defined via cells, or the PG via rules defined by `set_pg_via_master_rule` command. The `via_master_list` specifies the list of via masters. Via masters are created at the intersection center without replication. If offset or replication is needed, via rules specified by `set_pg_via_master_rule` command can be used. The `via_master_list` can also contain a list of PG via rules. `NIL` can be used as a keyword to indicate that no via is created. `default` can be used to describe the default vias in the specified location. If `NIL` is not specified in the list, the default contact code is used to complete a stacked via or to replace a via with DRCs when applicable.

By default, the default via defined in the technology file is

created at each intersection of orthogonal connection straps and macro pins in different layers.

`-pin_layers layer_list | @var`

Specifies a list of layers for allowed pin connections. Only pins on the specified layers can be used for a PG connection. If not specified, pins on all layers will be considered for a PG connection.

DESCRIPTION

Creates a macro power ground connection pattern. This pattern can be instantiated in a design with the `set_pg_strategy` command to connect macros to the power ground network. There are three types of macro connections: `long_pin`, `ring_pin` and `scattered_pin`. This pattern can be also applied to power pad connection.

EXAMPLES

The following example creates a macro connection pattern with long pin type. The pattern name is `long_pin_pattern`, the connection strap direction is horizontal, the layer is M5, symbolic net names are `pwr` and `gnd` respectively and width, spacing, pitch are specified through parameters `w`, `s` and `p` respectively. Default vias are created at all intersections between connection straps and macro pins.

```
prompt> create_pg_macro_conn_pattern long_pin_pattern \  
-pin_conn_type long_pin -nets {pwr gnd} \  
-direction horizontal -width @w \  
-layers M5 -spacing @s -pitch @p \  
-parameters {w s p}
```

The following example creates a macro connection pattern with ring pin type. The pattern name is `ring_pin_pattern`. The horizontal and vertical layers are M5 and M6 respectively. Width, spacing, pitch are specified by parameters.

```
prompt> create_pg_macro_conn_pattern ring_pin_pattern \  
-pin_conn_type ring_pin \  
-width {@hor_w @ver_w} -layers {M5 M6} \  
-spacing {@hor_s @ver_s} -pitch {@hor_p @ver_p} \  
-parameters {hor_w ver_w hor_s ver_s hor_p ver_p}
```

The following example creates a macro connection pattern with scattered pin connection type. The pattern name is `scatter_pin_pattern`. The connection strap width is specified with parameters. Horizontal and vertical layers are M5 and M6 respectively. Default vias are created between macro pins and connection straps.

```
prompt> create_pg_macro_conn_pattern scatter_pin_pattern \  
-pin_conn_type scattered_pin -nets {pwr gnd} \  
-width {@hor_w @ver_w} -layers {M5 M6} \  
-parameters {hor_w ver_w}
```

MORE EXAMPLES

For more example, please start GUI and invoke the following command.
`gui_show task assistant -task "Design Planning:PG Planning->Examples->Overview"`.

SEE ALSO

[compile_pg\(2\)](#)
[create_pg_composite_pattern\(2\)](#)
[create_pg_ring_pattern\(2\)](#)
[create_pg_std_cell_conn_pattern\(2\)](#)
[create_pg_wire_pattern\(2\)](#)
[remove_pg_patterns\(2\)](#)
[report_pg_patterns\(2\)](#)
[report_pg_strategies\(2\)](#)
[set_pg_strategy\(2\)](#)
[set_pg_via_master_rule\(2\)](#)

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